SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO1: *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems.* *(BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12 **Total Marks:60**

Annexure A

CONCEPT MAPPING

Code of Conduct:

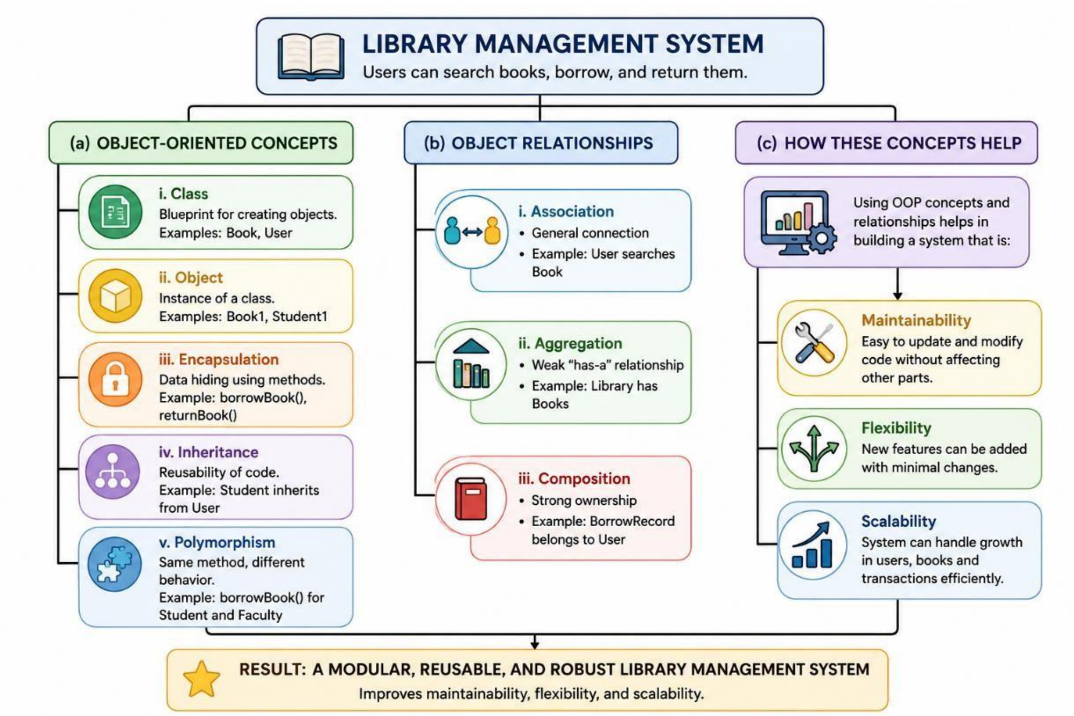
*I **Sayed Fazal (192311291)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Construct a Concept Map

GIVEN:

Requirement Analysis → _______→ Coding → Testing → _______→ Maintenance



Explanation

Requirement Analysis identifies the user's needs and system requirements. Design transforms these requirements into the software architecture. Coding converts the design into executable code. Testing verifies that the software functions correctly and meets requirements. Deployment releases the software to users, and Maintenance ensures continuous updates, bug fixes, and performance improvements after release.

Analysis

- Ensures a systematic Software Development Life Cycle (SDLC).
- Minimizes development risks by following sequential phases.
- Improves software quality, reliability, and maintainability.

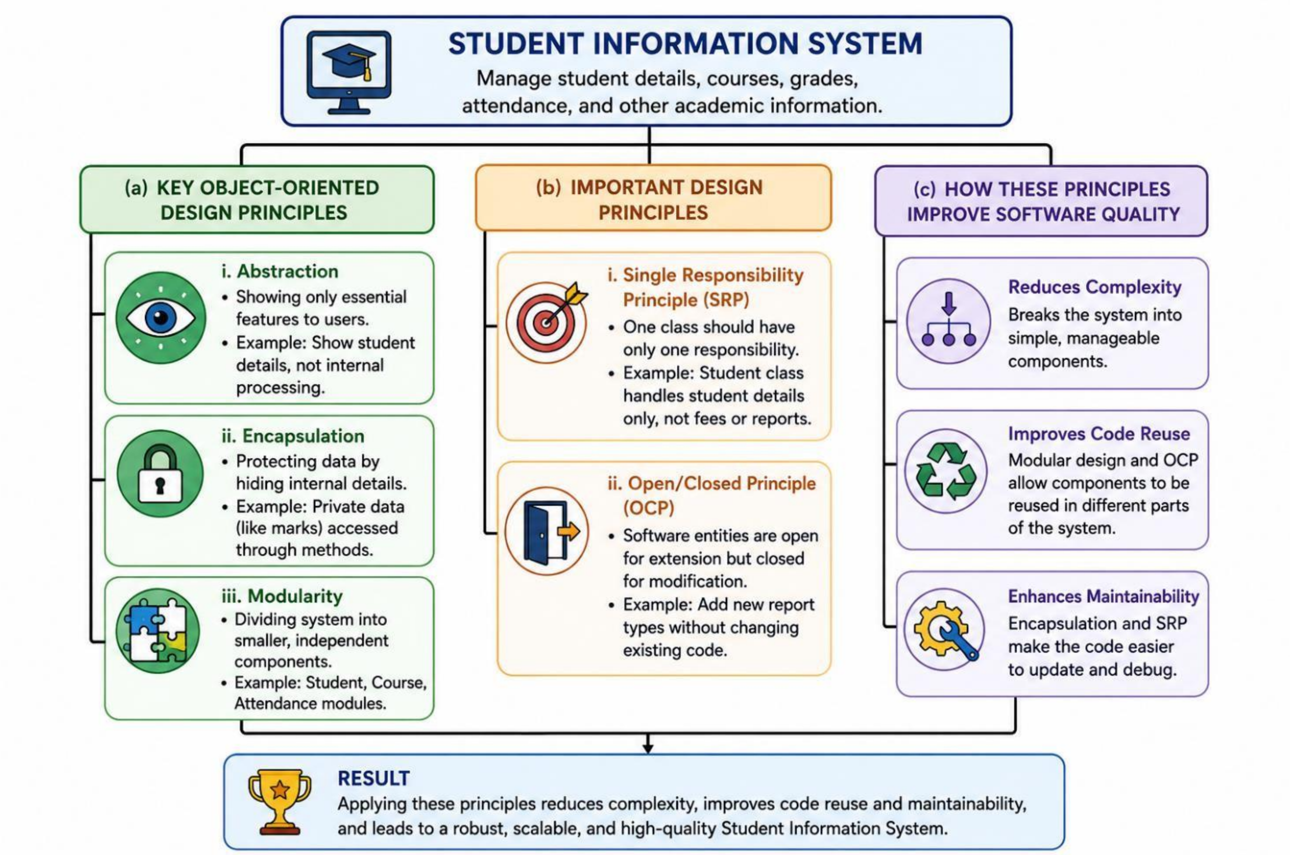
Object-Oriented Principles

→ Abstraction

→ ________

→ Inheritance

→ ________



Explanation

Abstraction hides unnecessary implementation details. Encapsulation protects data by restricting direct access. Inheritance enables a class to inherit properties and behaviors from another class. Polymorphism allows a single interface to perform different actions depending on the object.

Analysis

- Enhances security through data hiding.
- Promotes code reuse and flexibility.
- Simplifies software maintenance and extension.

3. Construct a Concept Map

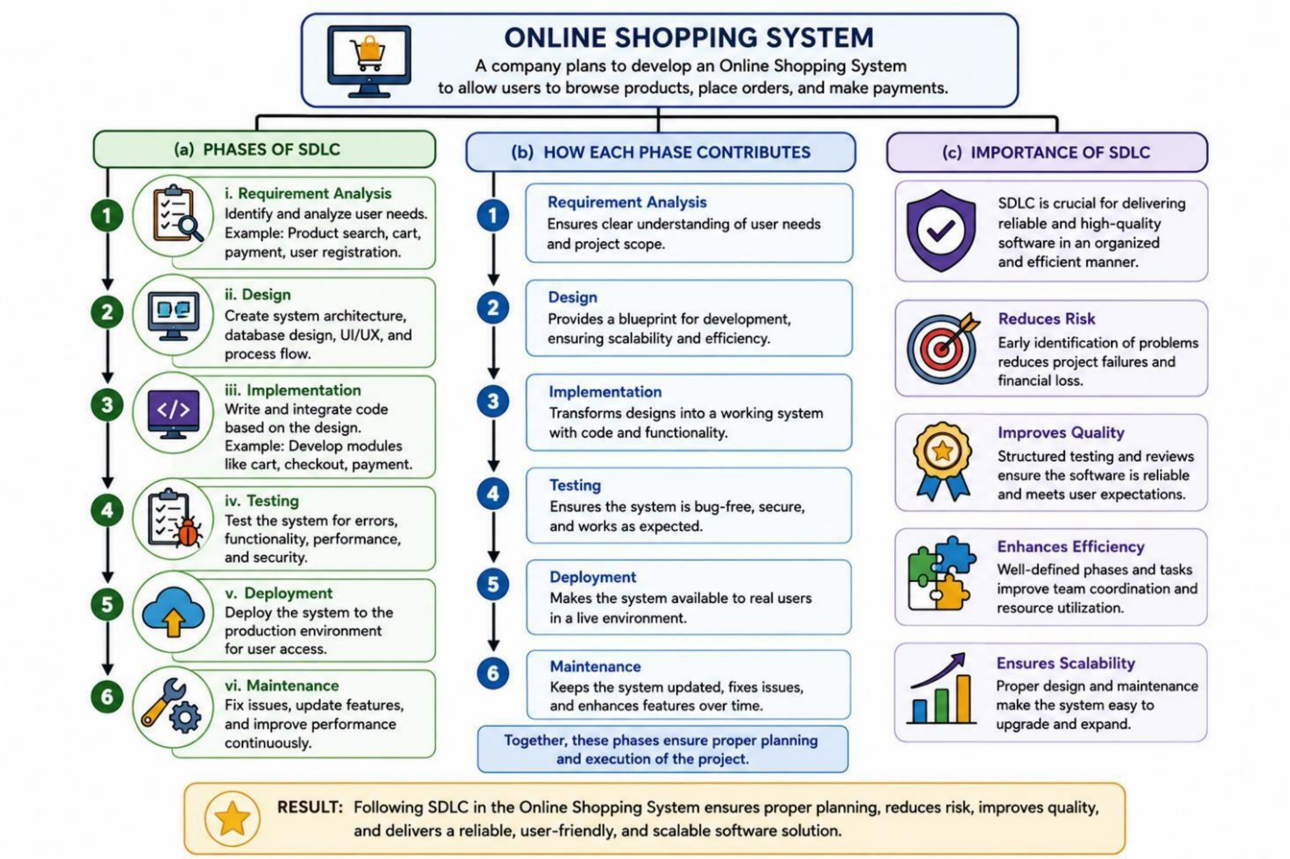
GIVEN:

Class → Blueprint

Object → Instance

Method → _______

Attribute → _______



Explanation

The Software Development Life Cycle (SDLC) is a structured process that includes Requirement Analysis, Design, Implementation, Testing, Deployment, and Maintenance for systematic software development.Each phase contributes to proper planning, organized execution, and efficient project management, ensuring that the software is developed in a controlled manner.

Analysis:

- Represents real-world entities effectively.
- Clearly separates data and functionality.
- Supports object-oriented system design.

4. A Concept Map Shows

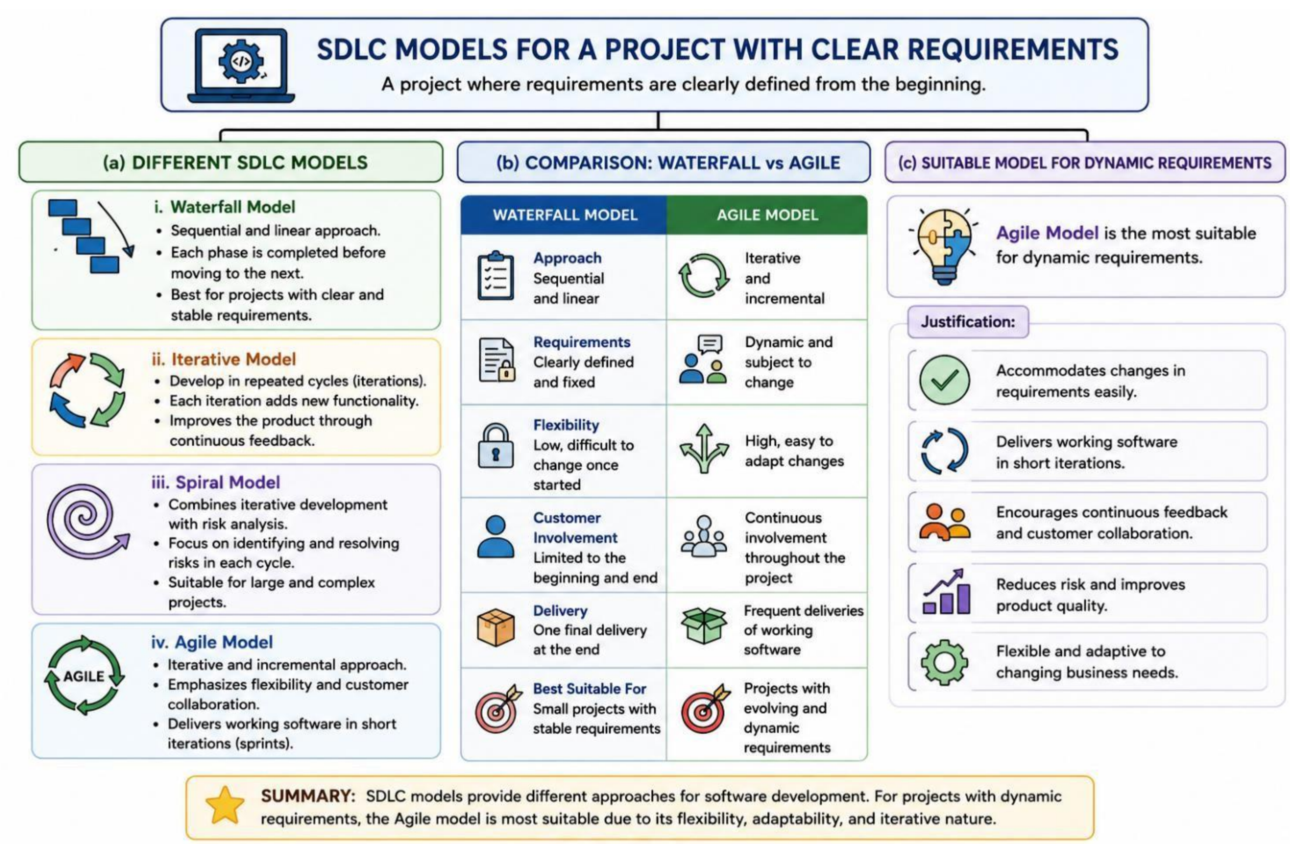
GIVEN:

Software Quality

→ Reliability

→ _______

→ Maintainability



CONCEPT MAP

Explanation

The Software Development Life Cycle (SDLC) offers different development models such as Waterfall, Iterative, Spiral, and Agile, each designed to meet different project requirements. The Waterfall model follows a sequential and rigid approach, whereas the Agile model is flexible, iterative, and supports continuous feedback and frequent updates.

Analysis

- Improves user satisfaction.
- Increases software acceptance.
- Supports long-term software success.

5. UML Concept Map

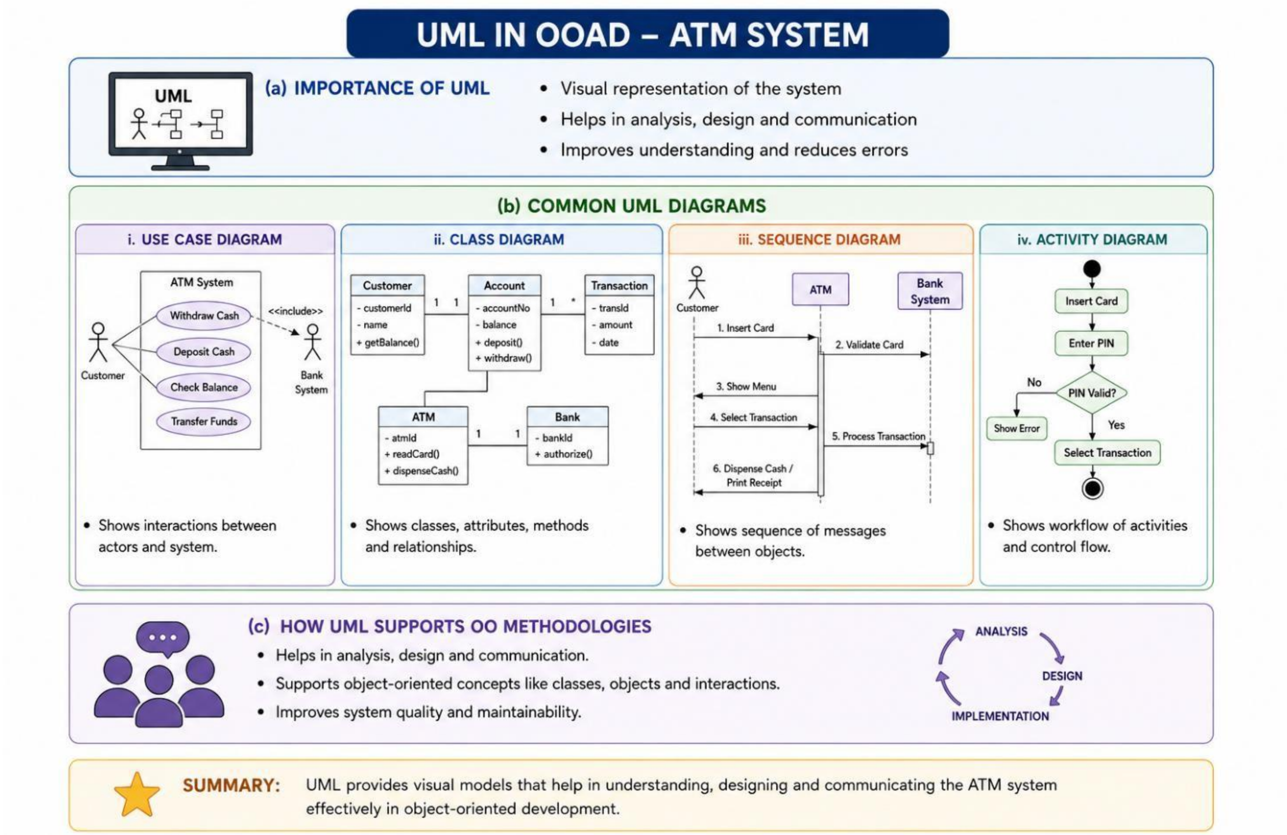
GIVEN:

UML

→ Structural Diagrams → _______

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → _______



Explanation

The Unified Modeling Language (UML) provides a visual representation of an ATM System using diagrams such as Use Case, Class, Sequence, and Activity diagrams to model system structure and behavior. These UML diagrams help analyze user interactions, define classes and relationships, illustrate object communication, and represent the workflow of ATM operations.

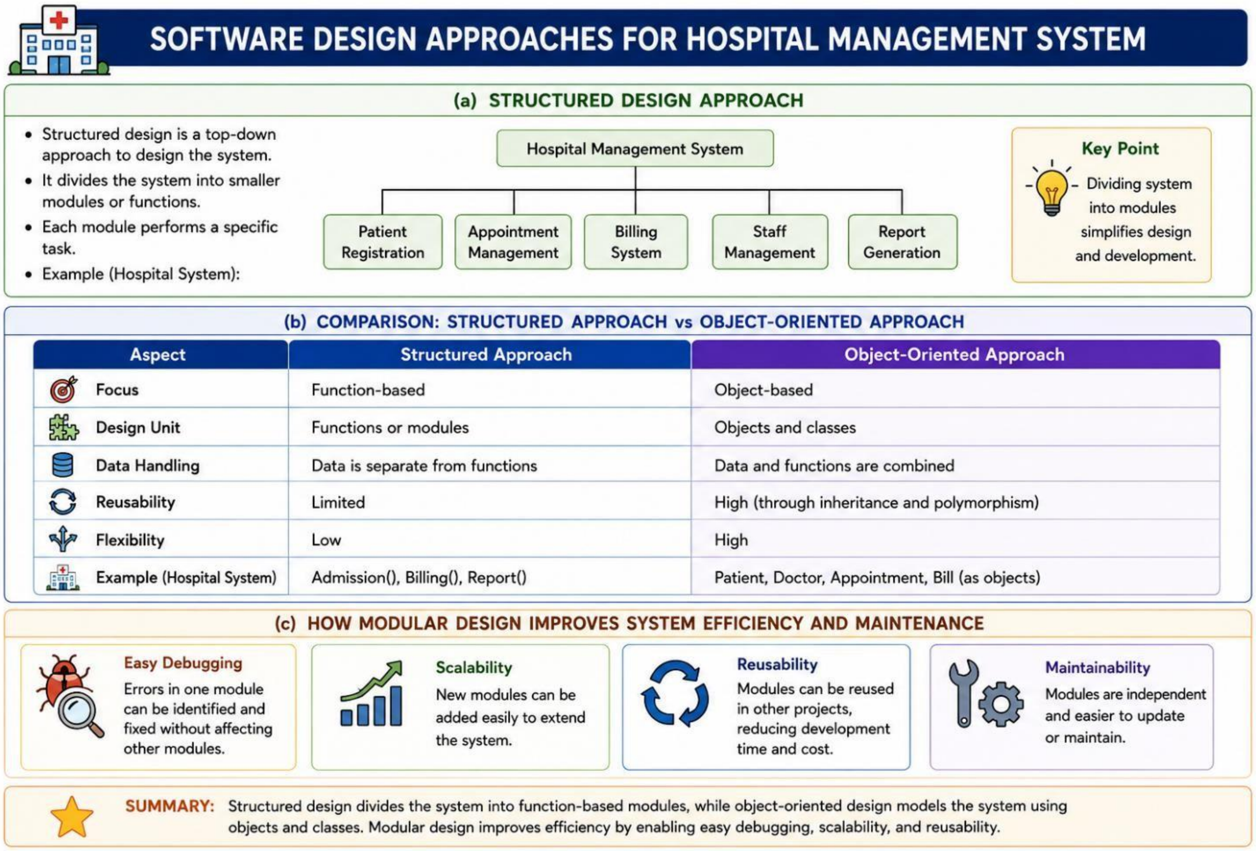
Analysis

- Improves visualization of software architecture.
- Supports object-oriented analysis and design.
- Enhances communication among stakeholders.

6. CONCEPT MAP

GIVEN:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗



Explanation

The structured design approach divides a Hospital Management System into smaller modules such as patient registration, billing, appointments, and reports, making development more organized and manageable.

The structured approach is function-based, whereas the object-oriented approach is object-based, improving flexibility, reusability, and real-world system modeling.

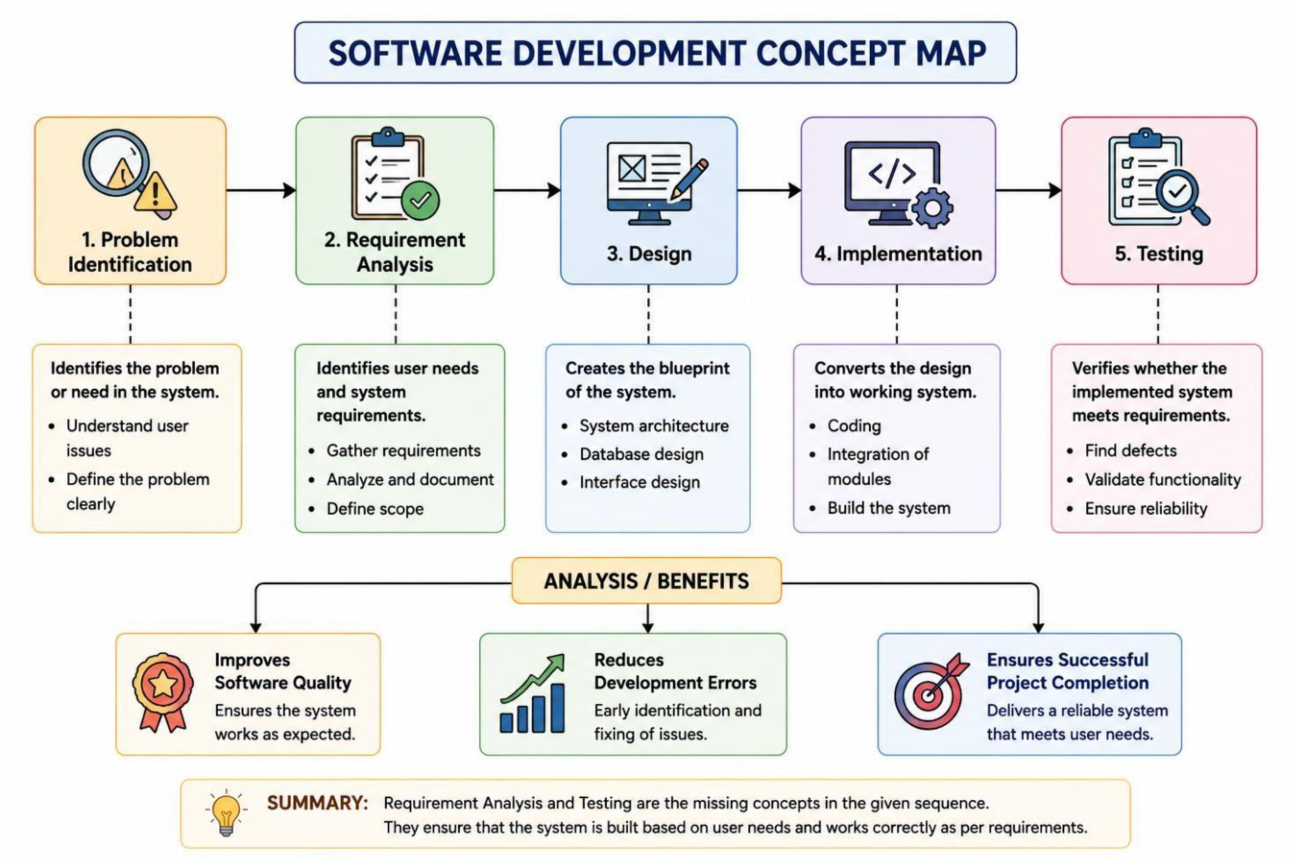
Analysis

- Prevents effective testing.
- Delays project completion.
- Reduces software reliability.

7. Construct a Concept Map

GIVEN:

Problem Identification → _______→ Design → Implementation → _______



Explanation

Problem Identification defines the project objective. Requirement Analysis gathers user requirements. Design creates the software architecture. Implementation develops the software, and Testing verifies whether it satisfies functional and non-functional requirements.

Analysis

- Reduces development errors.
- Improves software quality.
- Ensures successful project completion.

8. Construct a Concept Map

GIVEN:

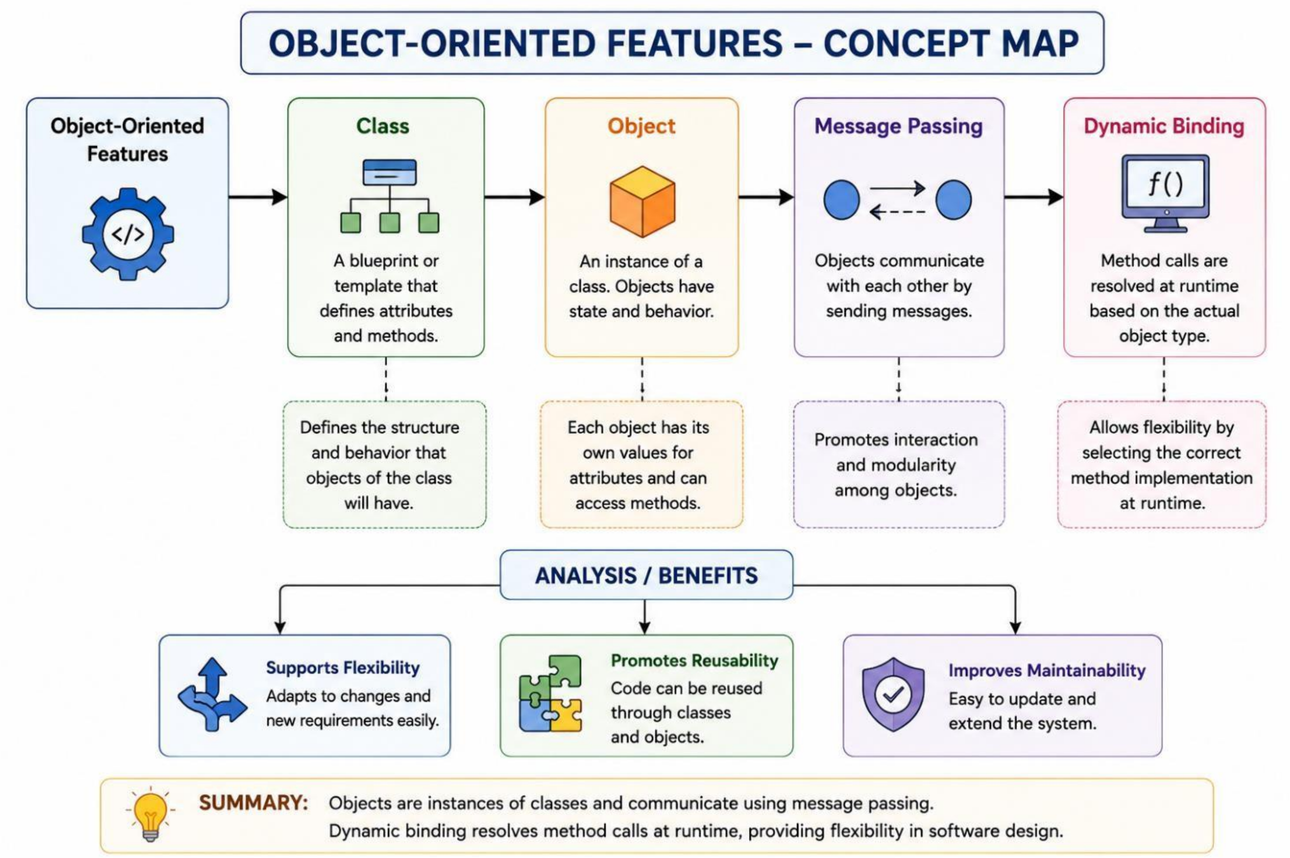
Object-Oriented Features

→ Class

→ _______

→ Message Passing

→ _______



Explanation

A Class defines the structure of objects. Objects communicate through message passing. Dynamic Binding determines the method implementation during runtime, enabling runtime polymorphism.

Analysis

- Improves flexibility.
- Enhances maintainability.
- Supports efficient software design.

9. Construct a Concept Map

GIVEN:

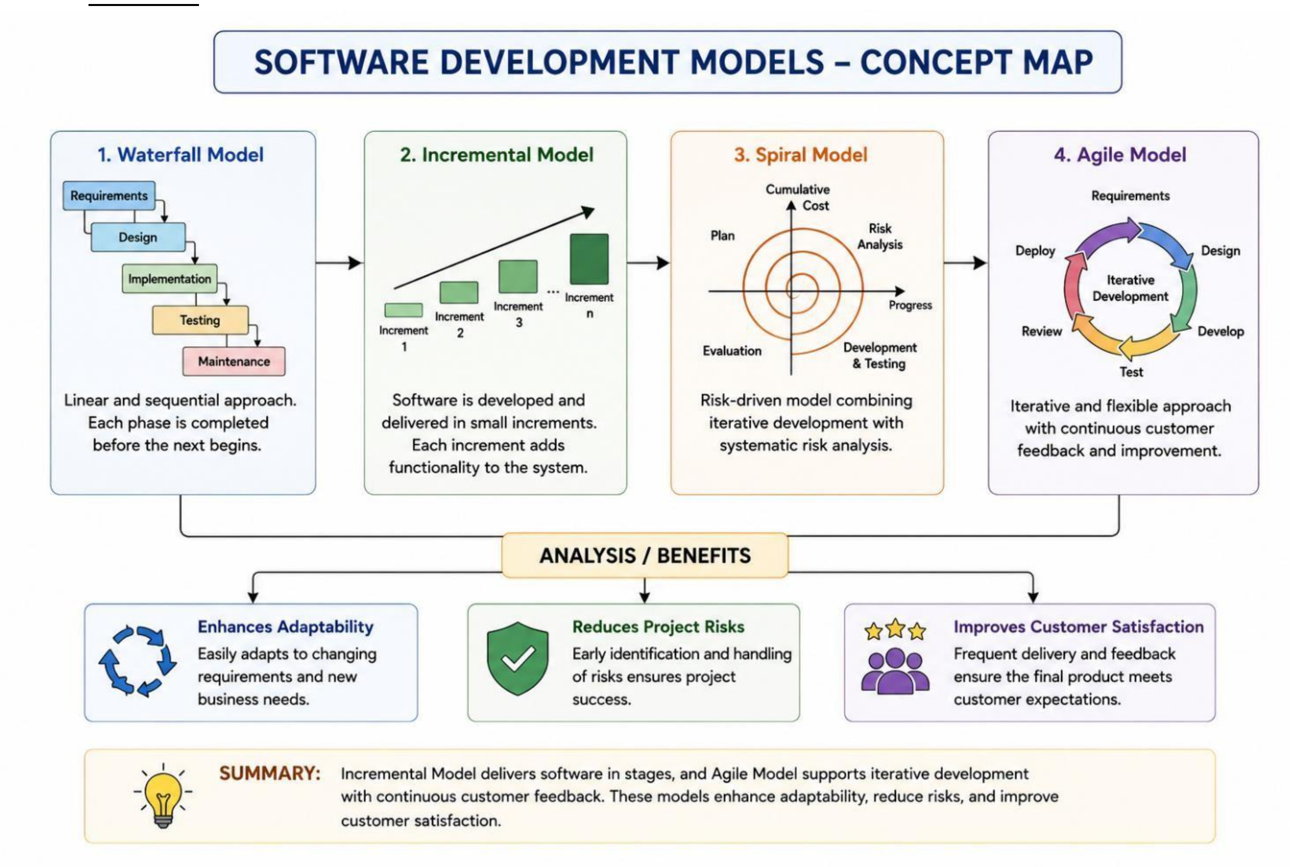
Software Development Models

→ Waterfall

→ _______

→ Spiral

→ _______



Explanation

The Waterfall Model follows a sequential process. The Incremental Model delivers software in stages. The Spiral Model combines iteration with risk management. Agile emphasizes iterative development and continuous customer collaboration.

Analysis

- Adapts to changing requirements.
- Reduces project risks.
- Increases customer satisfaction

10. Construct a Concept Map

GIVEN:

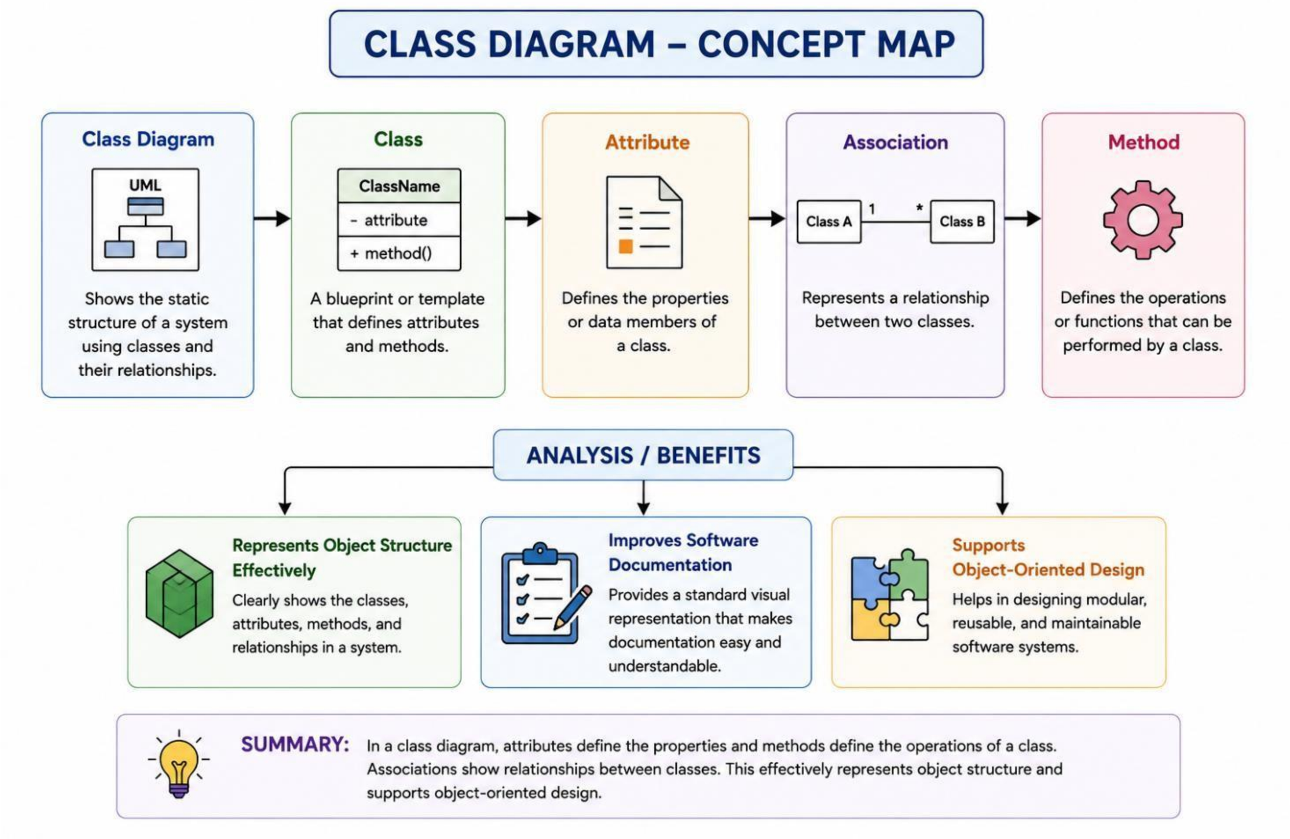
Class Diagram

→ Class

→ _______

→ Association

→ _______



Explanation

Classes represent system entities. Attributes define the characteristics of classes. Methods specify operations performed by classes. Associations describe relationships between classes.

Analysis

- Models object-oriented systems effectively.
- Improves documentation.
- Supports software implementation.

11. Construct a Concept Map

GIVEN:

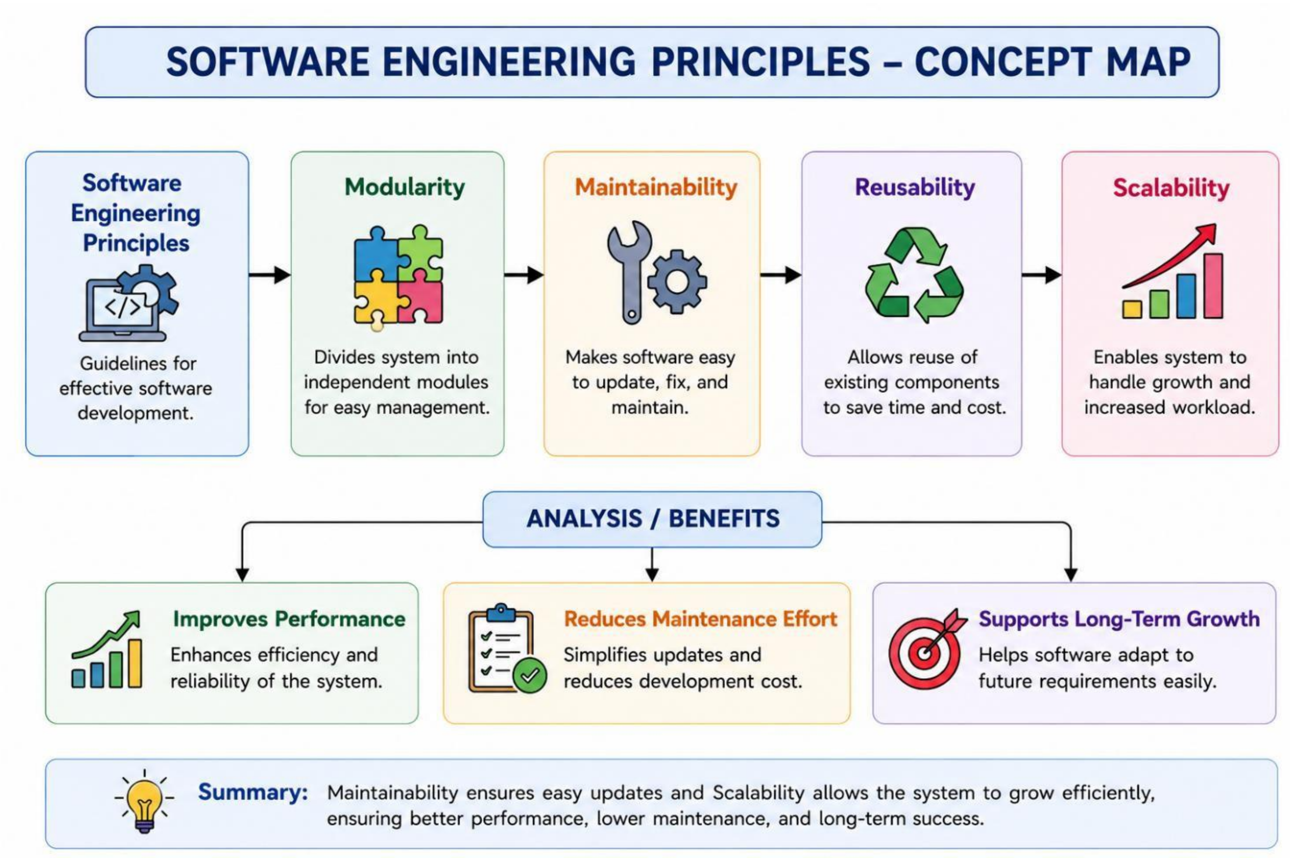
Software Engineering Principles

→ Modularity

→ _______

→ Reusability

→ _______



Explanation

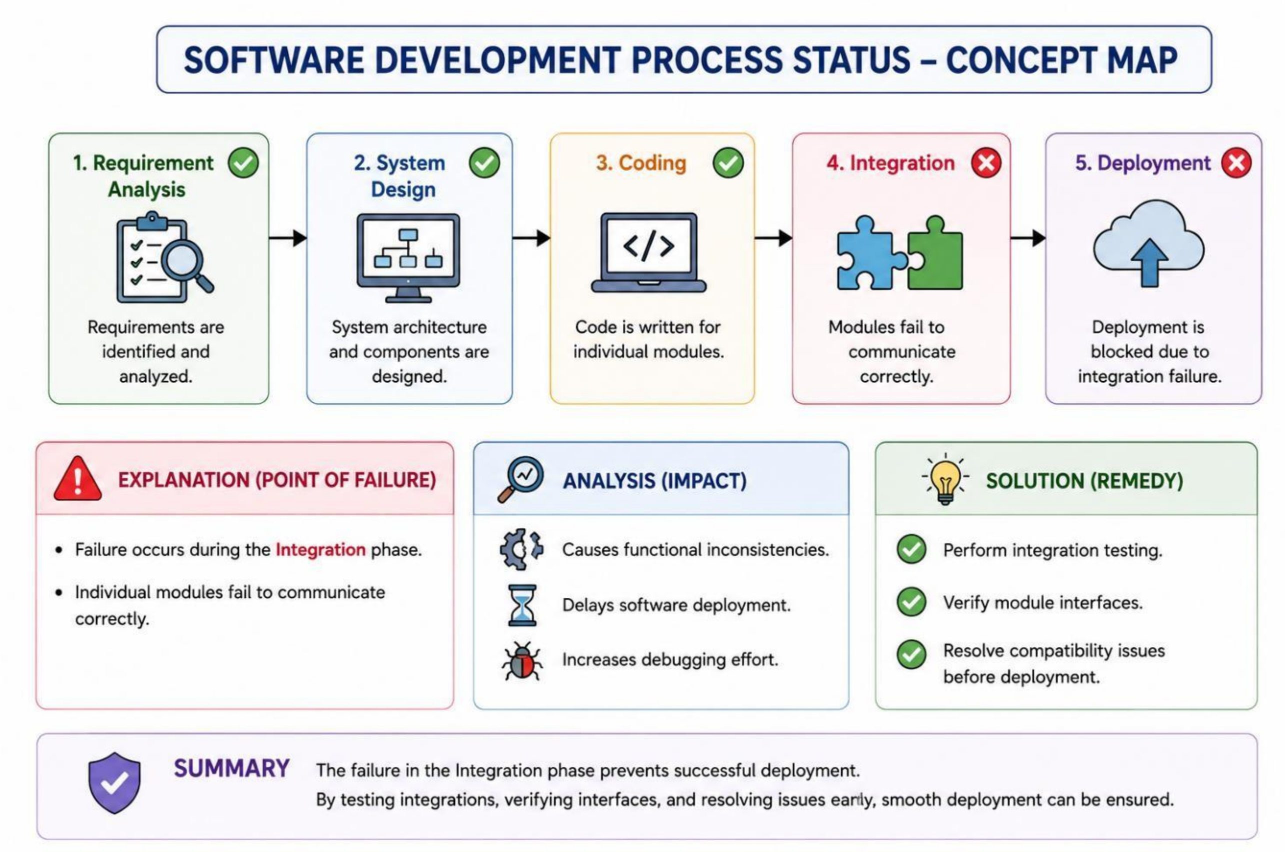
Modularity divides software into manageable components. Maintainability allows easy modification. Reusability enables components to be used in multiple projects. Scalability ensures the system performs efficiently as demand increases.

Analysis

- Reduces maintenance cost.
- Improves system performance.
- Supports future software growth.

12. GIVEN:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✕ → Deployment ✕



Explanation

The failure occurs during the Integration phase because independently developed modules fail to interact correctly due to incompatible interfaces or communication issues.

Analysis

- Produces functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand